Justin Jaffeth Corea Masís

Electronic Engineering

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Profile

Electronic Engineer with strong foundation in embedded systems, digital design, computer architecture and machine learning. Passionate about embedded systems and eager to contribute to safety-critical products while continuing to grow in software engineering and verification practices.

Projects

16-bit RISC-V Processor Design (Verilog) *Computer Architecture (2024)*

- Designed and simulated a 16-bit RISC-V microarchitecture using Verilog.
- Designed a pipelined 16-bit RISC-V processor in Verilog including ALU, register file, hazard detection and control logic.
- Developed testbenches in assembly to verify instruction execution and pipeline behavior.

Networked Sensor Node *Data Science (2026)*

- Developed firmware in MicroPython for Raspberry Pi Pico W sensor nodes.
- Designed a lightweight TCP communication protocol for automatic node registration and telemetry.
- Implemented end-to-end data acquisition system integrated with Python server and MySQL database.
- Used persistent storage to maintain device identity across reboots.

DC Motor Speed Control System (12V) *Hardware Design (2025)*

- Designed and implemented a closed-loop motor speed control system using analog circuitry and sensor feedback.
- Developed hardware for real-time speed regulation under variable load conditions.
- Integrated user interface including emergency stop, reset, mode selection (manual/automatic), and speed visualization.
- Implemented signal conditioning and control using operational amplifiers and sensor inputs.
- Validated system stability and response under different operating conditions.

Automatic License Plate Recognition System *Final Graduation Project (2025)*

- Designed and implemented an end-to-end computer vision pipeline (detection + OCR) for license plate recognition, targeting deployment in embedded systems for automated parking applications.
- Trained models using synthetically generated data to address dataset limitations, and validated performance on real-world data.
- Achieved 98.44% accuracy in plate recognition and 93.5% detection accuracy.
- Reached 3.84 FPS inference speed on resource-constrained hardware.

Education

Licenciate. Electronic Engineer *Instituto Tecnológico de Costa Rica (2020 – 2025)*
GPA: 87 / 100
Academic excellence scholarship

Technical Skills

Programming Languages: C/C++, Rust, Python
Embedded Systems: Bare-metal programming, STM32, Raspberry Pi, I2C/UART/SPI, UDP/TCP
Hardware Design: Verilog, Digital Design, Analog Design
Machine Learning: PyTorch, Computer Vision, Deep Learning, Model Optimization
Tools & Environments: Git, GDB, Linux, CMake, Docker, SQL

Languages

Spanish: Native | **English:** CEFR B2 (Prueba SpeakNow) | **Japanese:** JLPT N5